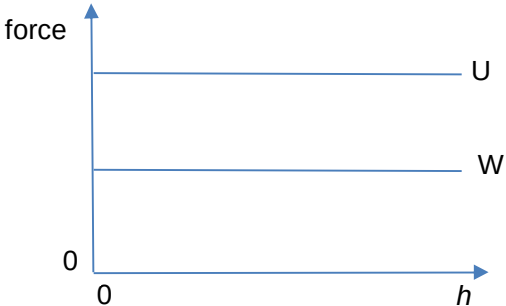
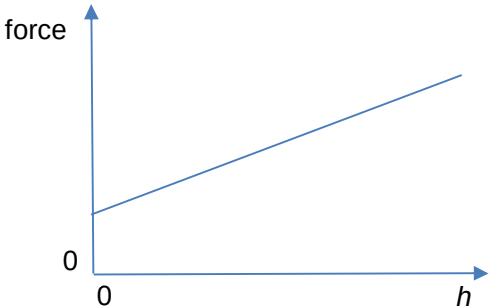


No	Solution
1(a)(i)	Upthrust on the sphere is the upward vertical force exerted by the liquid on the sphere due to the pressure difference between the upper and lower surfaces of the sphere. The pressure in the liquid increases with increasing depth ($p = h\rho g$). The pressure creates an upward force on the bottom surface of the sphere which is greater than the downward force created at its top surface. This pressure difference creates a resultant upward force on the sphere.
(a)(ii)	By Archimedes' Principle Upthrust = weight of the fluid displaced = mass of fluid displaced $\times$ acceleration of free fall = density of fluid $\times$ volume of liquid displaced $\times$ acceleration of free fall = $\rho_L Vg$ Since the sphere is fully submerged, the volume of the liquid displaced is equal to the volume of the sphere, V.
(b)	 The graph shows force on the vertical axis and depth h on the horizontal axis. The origin is marked with 0. Two horizontal lines are plotted: line U is higher than line W, indicating that the force represented by U is constant and greater than the force represented by W.
(c)	 The graph shows force on the vertical axis and depth h on the horizontal axis. The origin is marked with 0. A straight line starts at a positive value on the force axis and increases linearly with depth h.
2(a)	The atoms of ideal gas are in continuous random motion and have equal probability